RTH
CENTER FOR NON-CODING RNA IN TECHNOLOGY AND HEALTH

ENSAFE

UNIVERSITY OF COPENHAGEN

RIsearch3 for fast RNA-RNA interaction predictions

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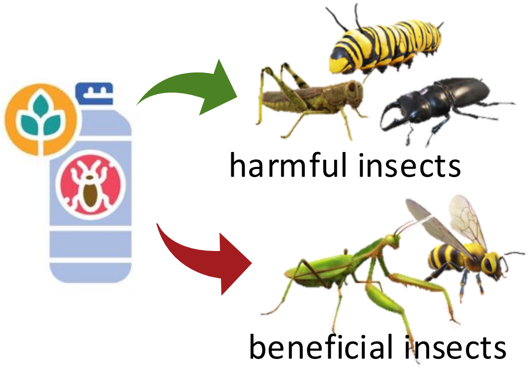
Background

- **Why does off-target assessment of biopesticides matter?**



Increased global food demand

Effective crop protection

Novel biopesticides

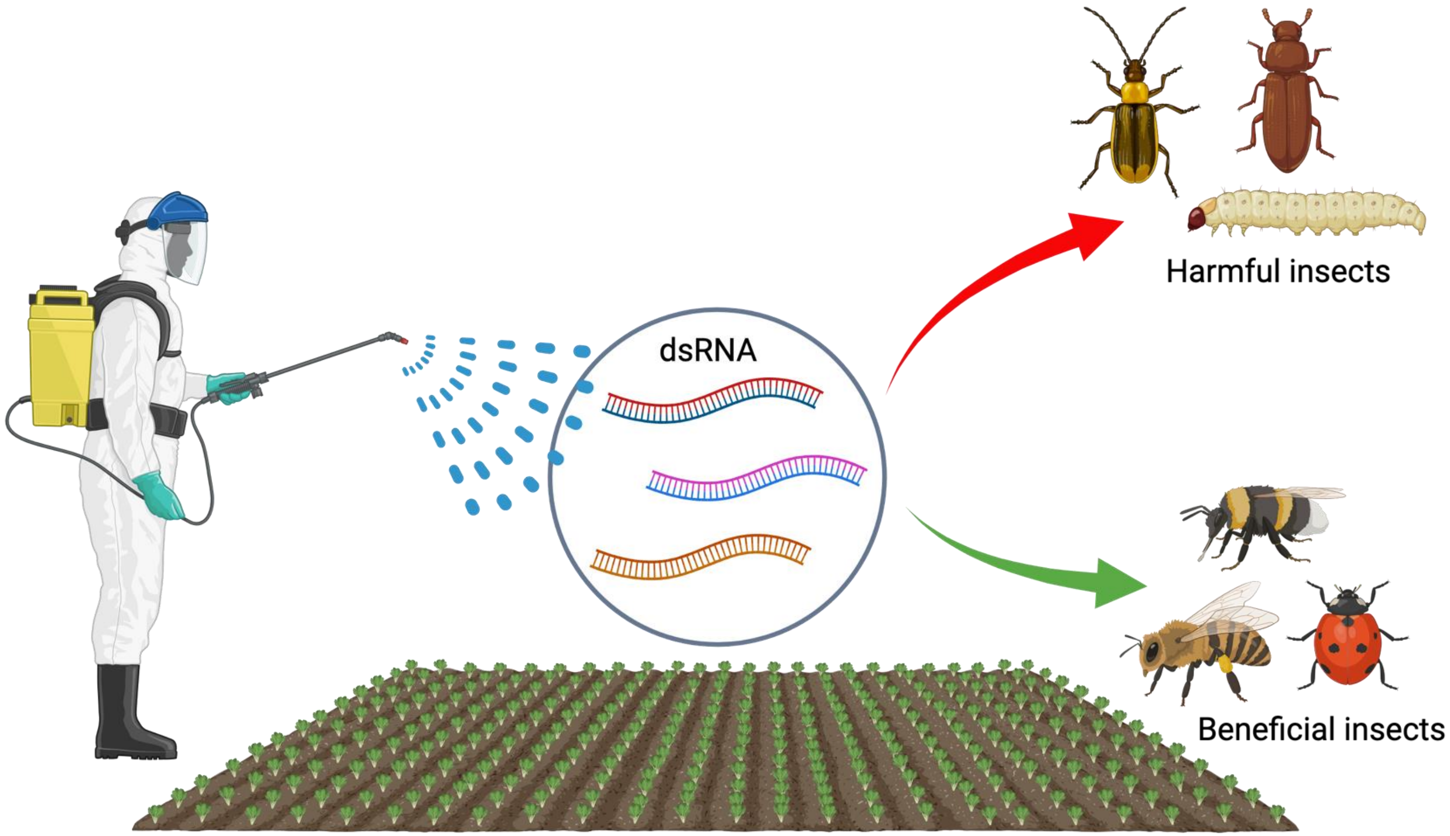
Lack of EU regulations

novo nordisk **foundation**

Novo Nordisk Foundation Challenge Grant no. NNF24OC0087112

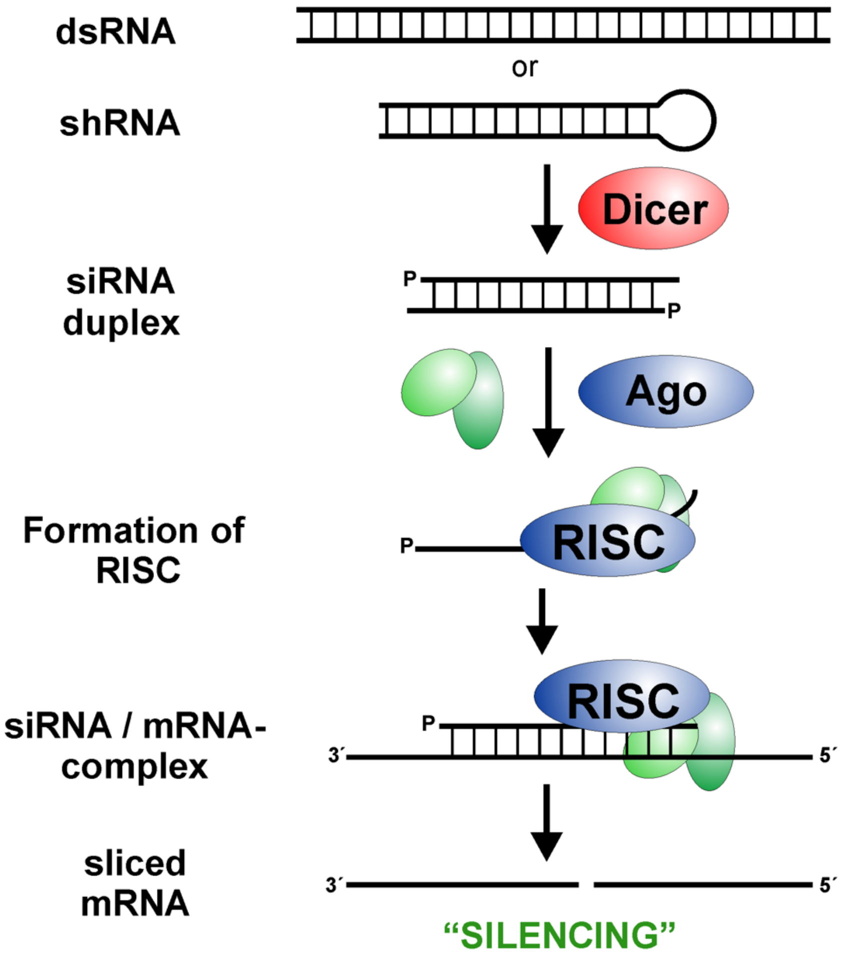
ENSAFE is an environmental risk assessment framework for siRNA- and peptide-based pesticides: integrating *in silico*, *in vitro*, and *in vivo* approaches.

How do we identify off-target organisms?

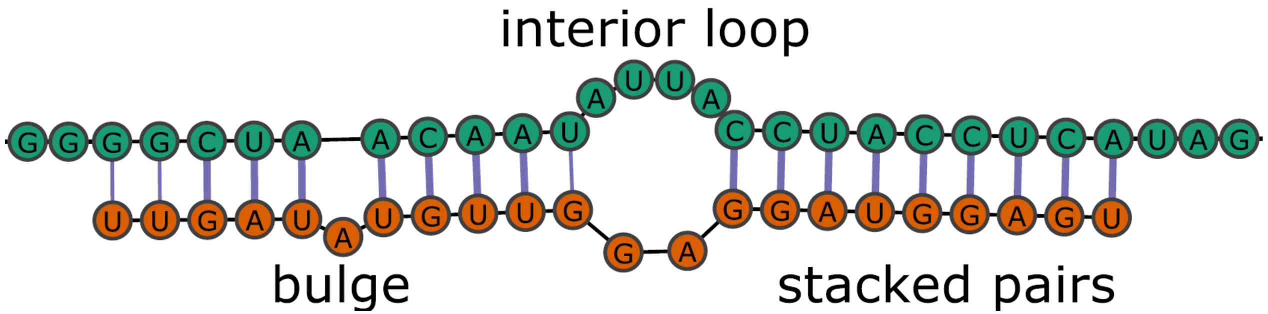


- *In silico* pipeline to quickly scan thousands of siRNAs through hundreds of genomes/transcriptomes
- Temperature-dependent RNA interactions

The complex scenario of RNAi



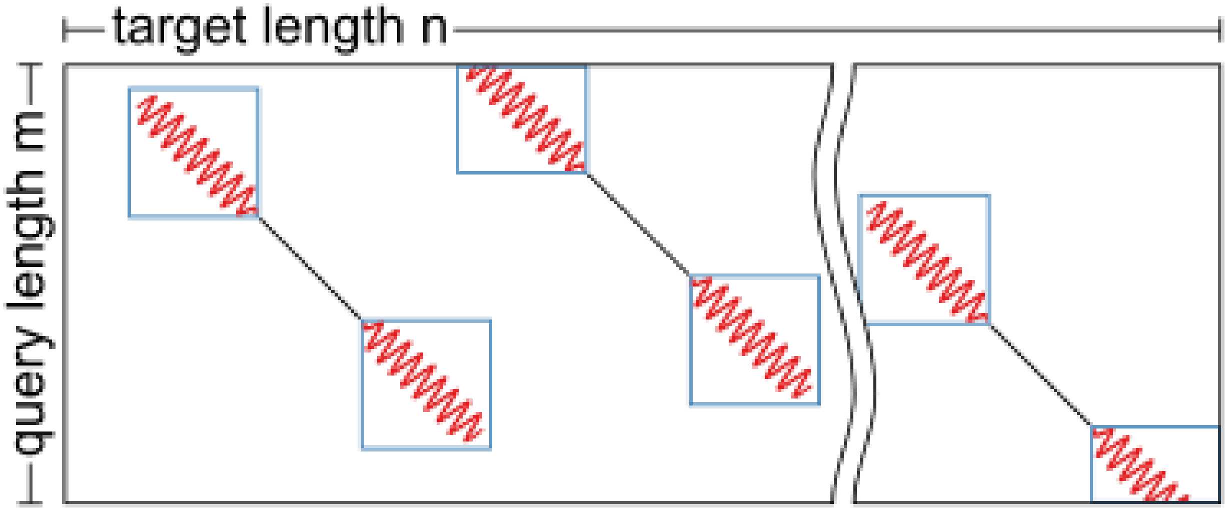
- Hundreds of genomes/transcriptomes → huge search space
- RNA binding is complex, complementary is not enough
- Purely thermodynamic algorithms scale poorly



http://www.uni-konstanz.de/FuF/chemie/jhartig/

RIsearch2

- <u>Seed-and-extend</u> strategy
- Suffix-array data structure for seed lookup (includes G-U wobble)
- Extends seeds on both ends with dynamic programming using a nearest neighbour energy model (simplified Turner) of RIsearch[†]
- Allows flexible parameter settings for different classes of RNA-RNA interactions
- <u>**Outputs raw binding energy**</u>



†Wenzel, Akbasli, Gorodkin, Bioinformatics, 2012.

siRNA off-target discovery pipeline

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Objective: To predict individual off-target transcripts and quantify the overall off-targeting potential (POFF) of an siRNA at a transcriptome-wide scale.

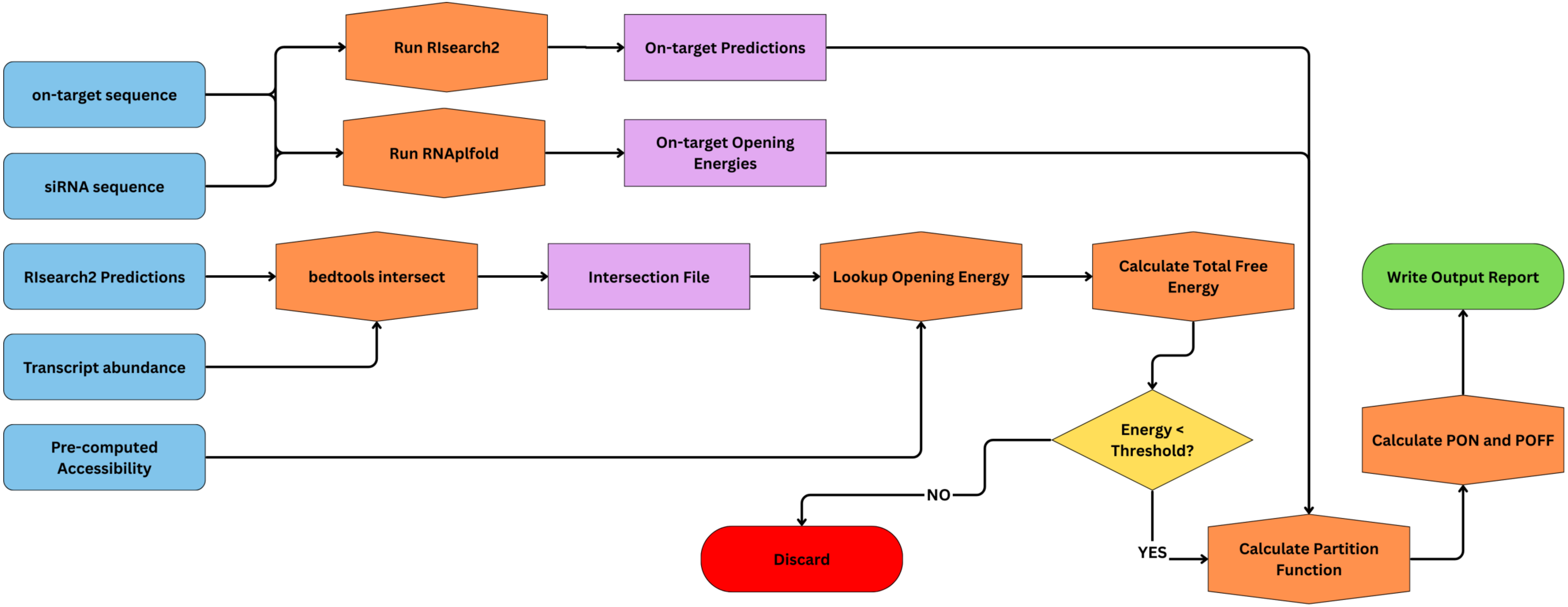
- Partition function[†] approach
- All putative siRNA-target interactions (RIsearch2[*]) represent a microstate variables
- State variables:
  - hybridization energy (RIsearch2)
  - opening energy (precomputed accessibilities with RNAplfold†)
  - abundance (expression data)

$$Z = \sum_{i \in \mathcal{I}} A_i e^{-\beta(E_i' + O_i)}$$

$$p_i = \frac{A_i e^{-\beta(E_i' + O_i)}}{Z}$$

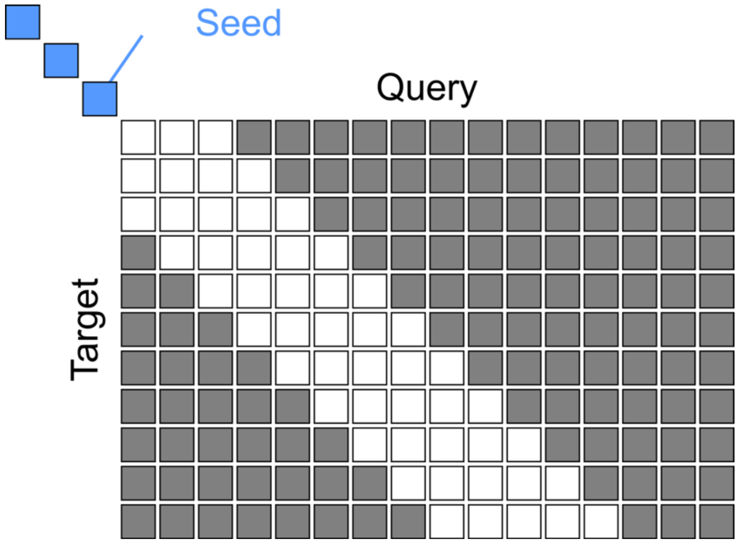
[†]The function of thermodynamic state variables describing the statistical properties of a system in thermodynamic equilibrium.
[*]Alkan et al., Nucl. Acids Res. 2017. †Bernhart ea, Bioinformatics, 2006.

Pipeline workflow



What's new in RIsearch3?

- **Temperature Dependence:** The tool now includes temperature-dependent energy parameters for **RNA/RNA**, **RNA/DNA**, and **DNA/DNA** interactions.
- Dedicated **CRISPR mode** with specific features for guide RNA (gRNA) target prediction: Bulged Seed Search, PAM Filtering, Repeat Filtering, CRISPRoff Scoring
- **Banded Alignments:** A "banded alignment" technique has been implemented to speed up dynamic alignment by only calculating values around the diagonal.



Work done by Christian Anton

New siRNA off-target pipeline implementation

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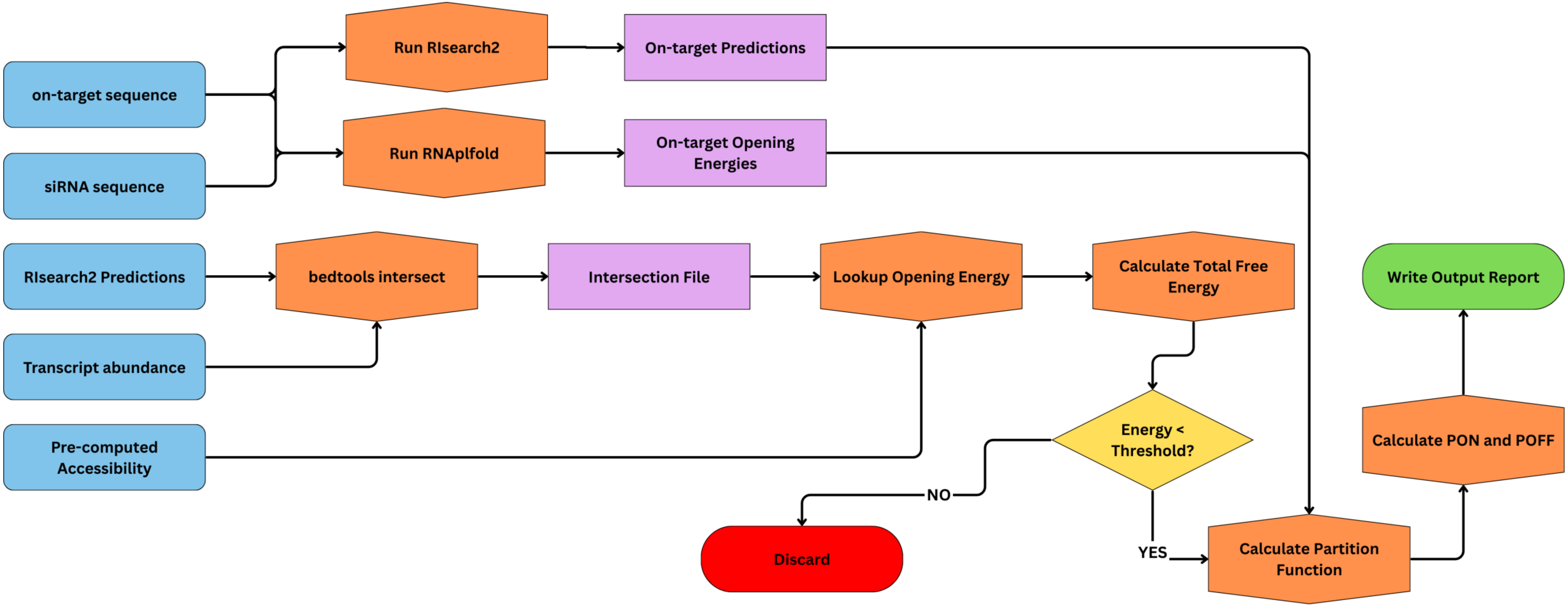
- Current: Python 3.14+, Polars, vectorized operations, lazy execution, memory-mapped accessibility profiles.
- Old: Python 2, heavy file I/O and shelling out (e.g., bedtools, sort, cut, RNAplfold, RIsearch2), manual parsing and lists in memory.

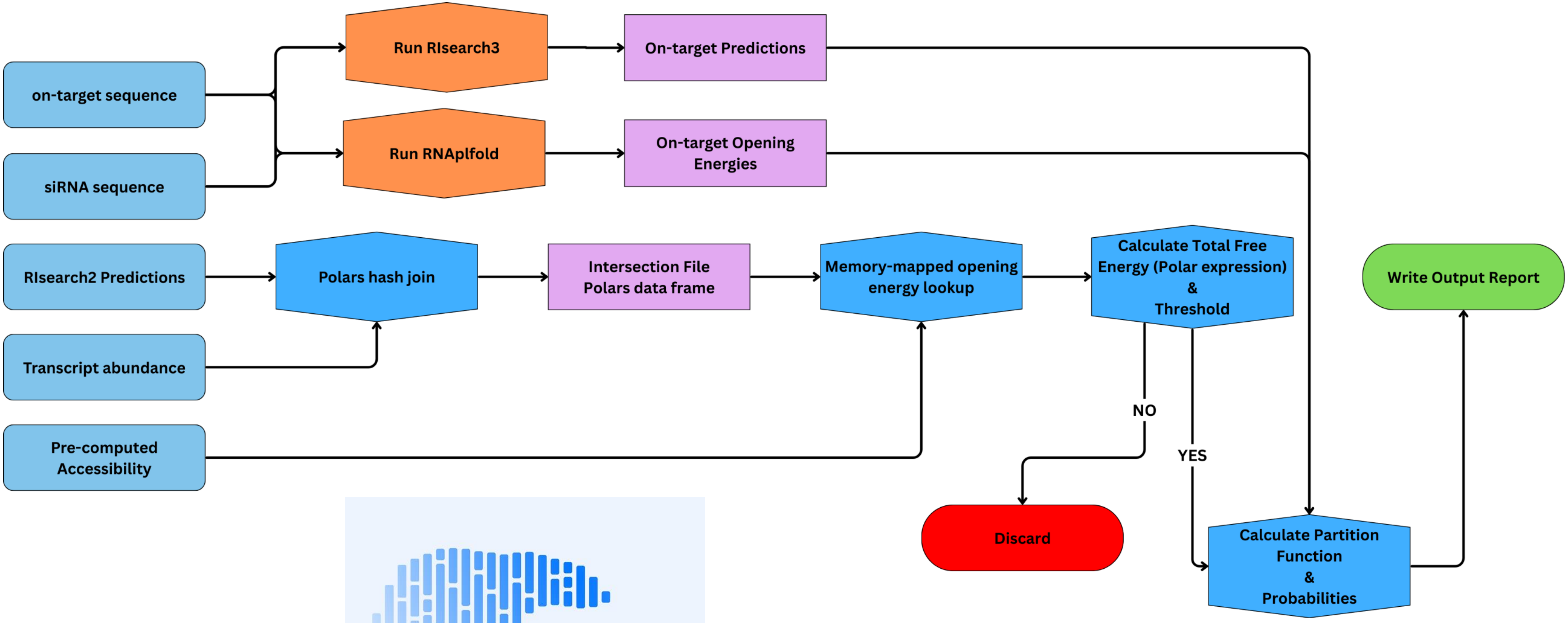
# siRNAs	Old	New	Change
100	104.553 s ± 1.099	52.231 s ± 0.370	~2 times faster 🚀
500	246.238 s ± 0.182	136.500 s ± 1.007	~2 times faster 🚀
1000	523.103 s ± 0.584	286.820 s ± 1.703	~2 times faster 🚀
2000	902.732 s ± 3.223	499.600 s ± 1.501	~2 times faster 🚀

ON-GOING WORK

Pipeline workflow



Pipeline workflow



New features

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- **Batch processing** for multi-siRNA workflows
- ‘uv’ package manager
- **Better maintainability** through service-oriented architecture
- **Modern CLI** with intuitive long-form flags and auto-generated help
- **YAML configuration** support for reproducible analyses

```
└─ risearch-pipeline --help

Usage: risearch-pipeline [OPTIONS] COMMAND [ARGS]...

siRNA off-target discovery pipeline – analyze RIsearch2 predictions.

╭─ Options ──────────────────────────────────────────────────────────────────
│ --config   -c      PATH  Path to YAML config file. Runs the command specified in the config.
│ --verbose  -v            Enable verbose logging (DEBUG level).
│ --help                   Show this message and exit.
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╭─ Commands ─────────────────────────────────────────────────────────────────
│ accessibility  Pre-compute accessibility profiles for a whole genome.
│ off-targets    Analyze siRNA off-target predictions.
│ orthologs      Download orthologs (OrthoDB) and transcriptomes (NCBI) for a list of species.
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ON-GOING WORK

What are the next steps?

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- RIsearch3 rust implementation
- RIsearch3 python bindings
- Implementation of ViennaRNA python API in the pipeline
- Further benchmark of banded version and temperature-dependent energy parameters

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- **Jan Gorodkin**

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